THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 13: Probability (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B\|A) = 0.4$, then $P(A \cap B)$ is: (A) 0.32 (B) 0.25 (C) 0.1 (D) 0.4	1
2.	The probability that a particular item is available in three shops A, B and C is $\frac{4}{5}$, $\frac{3}{4}$ and $\frac{2}{3}$ respectively. If a person visits all three shops, the probability that the item is available in at least one of them is: (A) $\frac{59}{60}$ (B) 1 (C) $\frac{1}{60}$ (D) $\frac{9}{60}$	1
3.	A die is thrown. If E is the event "the number is 4" and F is the event "the number is even", then $P(E\|F)$ is: (A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$	1
4.	If $P(A) = \frac{1}{2}$ and $P(B) = 0$, then $P(A\|B)$ is: (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) not defined	1
5.	A random variable X has the distribution $P(X = 0) = k$, $P(X = 1) = 2k$, $P(X = 2) = 3k$. The value of k is: (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 1	1

Q No.	Question	Marks
6.	**Assertion (A):** Events A and B with $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{6}$ are independent. **Reason (R):** Two events A and B are independent if and only if $P(A \cap B) = P(A)\,P(B)$. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Two cards are drawn one after the other without replacement from a well-shuffled pack of 52 cards. Find the probability that both are kings.	2
	LEVEL 2 **Q 8 – Q 14**	**18**
8.	The chances of P, Q and R getting selected as CEO of a company are in the ratio $4:1:2$. The probabilities that the company increases its profits under the new CEO P, Q or R are 0.3, 0.8 and 0.5 respectively. If the company increased its profits, find the probability that it is due to the appointment of R as CEO.	3
9.	Find the probability distribution of the number of heads in two tosses of a fair coin. Also find its mean.	2
10.	Given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$, find $P(A\|B)$, $P(B\|A)$ and $P(A'\|B')$.	3
11.	A and B independently try to solve a problem with probabilities $\frac{1}{2}$ and $\frac{1}{3}$ respectively. Find the probability that (a) the problem is solved (b) exactly one of them solves it.	3
12.	A family has two children. Find the probability that both are boys, given that at least one of them is a boy.	2
13.	A random variable X has $P(X = x) = kx$ for $x = 1,2,3,4$ and 0 otherwise. Find k, $P(X \geq 3)$ and the mean of X.	3
14.	Three coins are tossed. Let E: "three heads or three tails" and F: "at least two heads". Are E and F independent?	2
	LEVEL 3 **Q 15 – Q 20**	**25**
15.	A factory has machines A, B and C producing 50%, 30% and 20% of the items, of which 2%, 3% and 4% respectively are defective. An item picked at random is found defective. Find the probability that it was produced by machine A.	5
16.	In answering a multiple-choice question (4 options), a student either knows the answer or guesses. The probability that she knows the answer is $\frac{3}{4}$. If she answers correctly, find the probability that she knew the answer.	3
17.	Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of aces and its mean.	5

Q No.	Question	Marks
18.	A random variable X has the distribution: X: 0,1,2,3,4,5,6,7 with $P(X)$: $0,\ k,\ 2k,\ 2k,\ 3k,\ k^2,\ 2k^2,\ 7k^2+k$. Find k, $P(X<3)$ and $P(X>6)$.	3
19.	A laboratory blood test is 99% effective in detecting a disease when it is present, but gives a false positive for 0.5% of healthy persons. If 0.1% of the population has the disease, find the probability that a person who tests positive actually has the disease.	5
20.	**Case Study:** An NGO organises a charity event to distribute woollen caps to children in winter. The caps are in three boxes: Box I has 30 red caps, Box II has 20 red and 10 green caps, and Box III has 30 green caps. The probability that Box i is selected and a cap picked out is $\frac{i}{6}$, $i=1,2,3$. A person selects a cap. **(i)** What is the probability that he selects a red cap? **[2]** **(ii)** If he selects a green cap, what is the probability that it has come from Box II? **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 13: Probability

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) $0.8 \times 0.4 = 0.32$
2.	(A) $1 - \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{59}{60}$
3.	(C) $\frac{1/6}{1/2} = \frac{1}{3}$
4.	(D) $P(B) = 0$
5.	(A) $6k = 1$
6.	(A) $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$; R explains A.
7.	$\frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$
	LEVEL 2
8.	$P(E) = \frac{4}{7}(0.3) + \frac{1}{7}(0.8) + \frac{2}{7}(0.5) = \frac{3}{7}$; $P(R\|E) = \frac{1/7}{3/7} = \frac{1}{3}$
9.	$X = 0,1,2$ with $P = \frac{1}{4}, \frac{1}{2}, \frac{1}{4}$; mean $= 1$
10.	$P(A\|B) = \frac{2}{3}$, $P(B\|A) = \frac{1}{3}$, $P(A'\|B') = \frac{1-0.7}{0.7} = \frac{3}{7}$
11.	(a) $1 - \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{3}$ (b) $\frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2}$
12.	$\frac{1/4}{3/4} = \frac{1}{3}$
13.	$k = \frac{1}{10}$; $P(X \geq 3) = \frac{7}{10}$; mean $= \frac{1+4+9+16}{10} = 3$
14.	$P(E) = \frac{1}{4}$, $P(F) = \frac{1}{2}$, $P(E \cap F) = \frac{1}{8} = P(E)P(F)$; independent.
	LEVEL 3
15.	$P(D) = 0.01 + 0.009 + 0.008 = 0.027$; $P(A\|D) = \frac{0.010}{0.027} = \frac{10}{27}$
16.	$\frac{\frac{3}{4} \cdot 1}{\frac{3}{4} \cdot 1 + \frac{1}{4}\frac{1}{4}} = \frac{12}{13}$
17.	$X = 0,1,2$: $\frac{188}{221}, \frac{32}{221}, \frac{1}{221}$; mean $= \frac{34}{221} = \frac{2}{13}$
18.	$10k^2 + 9k = 1 \Rightarrow k = \frac{1}{10}$; $P(X < 3) = \frac{3}{10}$; $P(X > 6) = \frac{17}{100}$
19.	$\frac{0.001 \times 0.99}{0.001 \times 0.99 + 0.999 \times 0.005} = \frac{22}{133}$
20.	(i) $\frac{1}{6}(1) + \frac{2}{6}\left(\frac{2}{3}\right) + \frac{3}{6}(0) = \frac{7}{18}$. (ii) $P(\text{green}) = \frac{11}{18}$; $P(\text{II}\|\text{green}) = \frac{\frac{2}{6}\cdot\frac{1}{3}}{\frac{11}{18}} = \frac{2}{11}$